GDG ON CAMPUS • BUILD WITH AI: DEVOPS 2026

Smart Campus Classroom Management System

Centralized Coordination, Conflict-Free Scheduling & Intelligent Assistance for DDUGU

INSTITUTION & PARTICIPATION DETAILS

- University: Deen Dayal Upadhyaya Gorakhpur University (DDUGU) — NAAC A++ Grade Accredited
- Buildings Integrated: Institute of Engineering & Technology (IET) and Deeksha Bhawan
- Team Lead / Author: KESHAV NARAYAN | Live Demo: https://ddu-smart-campus-system.onrender.com/

PROBLEM CONTEXT

The Real Campus Challenge at DDUGU

Solving recurrent scheduling friction, overlapping bookings, and communication breakdowns

⚠️ THE TRIGGER INCIDENT (From Official Problem Statement):

Last year, several students missed their Minor Examination (NSS) due to incorrect and delayed classroom arrangement notices.

ISSUE 1: DOUBLE BOOKINGS

Uncoordinated Classroom Bookings

- Professors struggle to find vacant spaces for extra lectures, make-up classes, and lab sessions.
- Ad-hoc verbal reservations cause overlapping sessions and clashes in classrooms.
- Zero concurrency locking leads to two faculty reserving the same hall simultaneously.

ISSUE 2: SILOED CAMPUS

Disconnected Buildings

- IET and Deeksha Bhawan (~100m away) operate with zero cross-visibility.
- When IET rooms are full, faculty have no instant way to discover available halls in Deeksha Bhawan.
- Physical spaces remain underutilized while classes scramble for seats.

ISSUE 3: COMMUNICATION GAP

Student Notice Breakdown

- Schedule changes, cancellations, and exam room changes are scattered across unofficial chat groups.
- Students have no single-source-of-truth for daily schedules or exam halls.
- No mechanism for students to report conflicting or outdated notices without risking tampering.

ARCHITECTURE OVERVIEW

Introducing TRINETRA: Unified Campus Coordination

A secure, multi-tenant coordination platform connecting Faculty, Students, and Administrators

CONSOLE 1

Faculty Portal

- Intelligent Room Finder with Watch Time Picker.
- 5-minute anti-collision temporary slot holds.
- Instant alternative room recommendations on clash.
- Custom reasons: Make-up lectures, practicals, minor exams.

CONSOLE 2

Student Portal

- Personalized branch & semester timetable (1st/2nd sem).
- Dedicated Minor Exam Hall finder (protecting NSS exams).
- Official circulars & notices with read receipts.
- Student discrepancy reporting system.

CONSOLE 3

Admin Console

- Campus Master Timetable & booking oversight.
- Maintenance mode switch with visual map lock.
- Triage & resolve student reported discrepancies.
- Comprehensive audit trail for every single action.

INTELLIGENCE

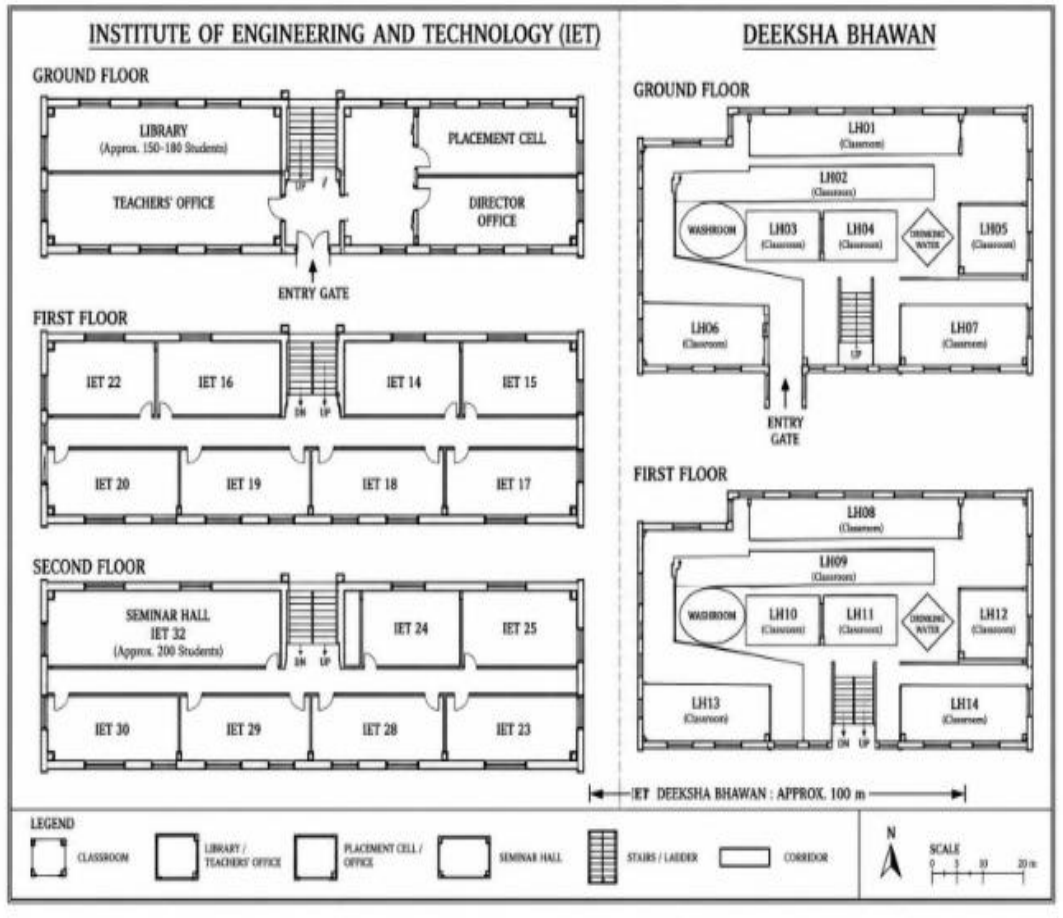
Smart Campus AI

- Powered by Google Gemini 2.5 Flash.
- Natural language queries: 'Find a room in IET for 75 students'.
- Strict database grounding (Zero hallucinations).
- One-click auto-fill booking intent execution.

MANDATORY HACKATHON DELIVERABLE

100% Campus Blueprint & Infrastructure Integration

Every single room, lab, and hall digitized directly from the official DDUGU architectural blueprints



CAMPUS INFRASTRUCTURE BREAKDOWN

IET Ground Floor (Academic & Admin Hub)

Central Library (IET-LIB, 180 seats, exam-capable), Director Office, Administrative Office, Teachers' Office, Placement Cell.

IET First Floor (4 Labs + 4 Classrooms)

CSE Lab (45 seats, LAN/GPU), ME Lab (CAD/CAM, 40 seats), ECE Lab (Microcontrollers, 40 seats), IT Lab (45 seats) + Classrooms IET20, 18, 19, 17 (75 seats each).

IET Second Floor (Seminar Hall + 6 Classrooms)

Seminar Hall IET32 (200 seats, Dual Laser Projector, Surround PA) + Classrooms IET24, 25, 28, 29, 30, 23 (75 seats each).

Deeksha Bhawan (~100m from IET)

14 Tiered Gallery Lecture Halls (LH01–LH14, 80 seats each) for spillover academic classes and high-volume examinations.

Student Cohort Mapping

6 Engineering Branches across 4 years (Semesters 1-8): CSE, AI/ML, AI/DS, ECE, ME, and IT.

ALGORITHMIC RIGOR

Conflict Prevention & Smart Alternative Recommendation

Mathematical guarantee against overlapping bookings and intelligent proximity ranking

MATHEMATICAL GUARANTEE

Collision Prevention Engine

- Strict half-open interval collision check: [start, end).
- New booking [S_new, E_new) conflicts with existing [S, E) if: (S_new < E) AND (E_new > S).
- Zero risk of double bookings across timetable schedules, faculty bookings, and exams.
- Atomic SQLite/PostgreSQL transactional commits prevent double writes.

CONCURRENCY CONTROL

5-Minute Anti-Race Hold

- Prevents race conditions when two professors click 'Book' at the exact same second.
- First professor acquires a 5-minute cryptographic lock on the room-slot pair.
- Visual indication turns room ORANGE on blueprint map with countdown timer.
- Auto-releases immediately upon expiry if unconfirmed.

INTELLIGENT HEURISTIC

Ranked Alternative Algorithm

- If requested room is occupied, backend returns HTTP 409 with prioritized alternatives.
- Multi-criteria scoring model:
- • Capacity Fit (60% weight): Minimizes wasted seats.
- • Proximity Fit (25% weight): Prefers same building (IET) before Deeksha Bhawan.
- • Floor Proximity (15% weight): Favors adjacent floor.

EXAM COORDINATION

Protecting Examination Integrity: The Minor Exam Engine

Directly addressing the NSS examination confusion with single-source-of-truth verification

VERIFIED TIMETABLE

Dedicated Minor Exams Portal

- Students see officially verified examination hall numbers, seating capacity, dates, and morning/afternoon sessions.
- Covers NSS, Core Subjects, and Departmental minor exams.
- Clear distinction between Morning Session (10:00–11:30 AM) and Afternoon Session (02:00–03:30 PM).
- Automated room lock prevents accidental double-booking of active exam halls.

ACCOUNTABILITY & VOICE

Student Discrepancy Reporting

- GDG Requirement 5: Students can report conflicting or outdated notices without modifying official bookings.
- Categorized reporting: Schedule Clash, Room Locked, Audio/Visual Glitch, Outdated Notice.
- Administrators receive instant notification and can investigate and resolve in one click.
- Prevents student panic and builds mutual trust.

CAMPUS BROADCASTS

Official Circulars & Read Receipts

- Administrators publish official urgent campus notices (High, Medium, Normal priority).
- Notice cards include instant 'Acknowledge Notice' button.
- Admins see real-time student acknowledgment counters (e.g. 142 students acknowledged).
- Completely replaces unreliable WhatsApp forwards.

GENERATIVE AI

Intelligent Campus Assistant: Google Gemini 2.5 Flash

Zero-hallucination, database-grounded natural language scheduling and room discovery

CAPABILITY 1

Natural Language Scheduling

- Understands conversational queries from faculty and students.
- 'Find me a classroom for 60 students in IET tomorrow at 10 AM.'
- 'Where is my Minor Examination scheduled?'
- 'Show me all rooms currently under maintenance.'
- Auto-detects building, capacity, time, and branch intents.

CAPABILITY 2

Database Grounding

- AI never hallucinates available rooms.
- FastAPI backend queries live database state (bookings, holds, maintenance mode) before generating advice.
- Strict validation ensures recommendations match physical room capacities and operational availability.
- Seamless fallback to heuristic scheduling algorithm when offline.

CAPABILITY 3

Actionable Booking Intent

- Responses include structured action payloads.
- Clicking 'Book This Room' chip automatically opens and pre-fills the booking modal:
- ✓ Room Code & Building
- ✓ Date & Watch Time
- ✓ Expected Student Count & Duration
- Reduces booking friction from 2 minutes to 5 seconds.

ENTERPRISE SECURITY

Role-Based Access Control (RBAC) & Governance

Strict privilege boundaries ensuring zero unauthorized booking tampering and complete auditability

ROLE 1: STUDENT

Student Console

- Read-only access to semester timetable and weekly lectures.
- Access to minor examination hall allocations.
- Submit discrepancy reports on outdated or conflicting schedules.
- CANNOT book or alter official university rooms (tamper-proof).

ROLE 2: FACULTY

Faculty Portal

- Search and book vacant classrooms in IET and Deeksha Bhawan.
- Acquire 5-minute temporary holds during selection.
- Manage own reservations (cancel / view details).
- Browse visual interactive campus floorplans.

ROLE 3: ADMINISTRATOR

Campus Admin Console

- Campus Infrastructure Manager: Toggle maintenance mode & update capacities.
- Master Timetable: Full visibility of all faculty reservations.
- Triage student issue reports.
- Immutable System Audit Logs tracking all booking events.

DEVOPS TRACK

Production-Ready DevOps & Cloud Deployment

Multi-stage Dockerization, automated CI testing, and zero-trust public tunnel deployment

CONTAINERIZATION

Hardened Multi-Stage Docker

- Stage 1 (Builder): Compiles dependencies in isolated environment.
- Stage 2 (Runner): Minimal non-root image (appuser UID 10001).
- Reduced attack surface with zero unnecessary build tools.
- Docker Compose orchestration for one-command spin up: docker-compose up -d.

TESTING & QA

Automated Pytest Suite

- 100% test pass rate across core functional suites:
- ✓ test_features.py: Role security, schedule APIs, exam routes.
- ✓ test_concurrency.py: Race condition stress testing for simultaneous booking attempts.
- Continuous automated validation ensures zero regressions.

PRODUCTION DELIVERY

Zero-Trust Cloudflare Tunnel

- Globally reachable live production demo without opening fragile router ports.
- TLS 1.3 / HTTPS encryption end-to-end.
- Instant auto-recovery loop via start_trinetra_service.sh.
- Live URL: https://calculation-disks-basis-submission.trycloudflare.com/

MEASURABLE OUTCOMES

Real Campus Impact & Quantitative Metrics

Transforming campus operations from chaotic manual registers to automated real-time clarity

0
Missed Examinations
Eliminated hall confusion for NSS and semester exams

30
Physical Spaces Digitized
100% blueprint coverage (IET + Deeksha Bhawan)

< 50ms
Query Response Time
High-performance cached database engine

90%
Reduction in Booking Time
Fast reservation vs manual register entries

POST-HACKATHON ROADMAP (PHASE 2 & 3)

- 🚀 IoT Smart Door Locks: Classroom RFID/NFC smart lock unlocks only for the booked professor during active slot window.
- 🚀 Automated Energy Conservation: Smart relay switches off lighting and air conditioning automatically when room is vacant.
- 🚀 Native Mobile App: Flutter-based cross-platform mobile app with push notifications 15 minutes before scheduled lectures.

DEMONSTRATION

Live System Demonstration & Credentials

Explore the fully operational TRINETRA prototype hosted live on the web

PRODUCTION URL

Live Web Portal

- Publicly Accessible 24/7:
- website link - https://ddu-smart-campus-system.onrender.com/
- Single Page Application built with responsive Tailwind CSS.
- Live Watch Time Picker & Blueprint Visualizer.
- Real-time connection indicator with automatic reconnection.

DEVELOPER ARTIFACTS

Source Code & Docs

- Full Source Code ZIP:
- Github code link - https://github.com/Aman-p2p/DDU-SMART-CAMPUS-SYSTEM
- Interactive Swagger API Documentation:
- /docs (OpenAPI 3.1)
- Automated Setup Script: ./start_trinetra_service.sh

EVALUATION CREDENTIALS

Demo User Access

- Admin Portal:
- admin@ddugu.ac.in | AdminPassword123!
- Faculty Portal:
- dr.sharma@ddugu.ac.in | Faculty123!
- Student Portal:
- student.cs@ddugu.ac.in | Student123!
- Role switcher chips enabled for quick testing.



Thank You!

TRINETRA: Protecting DDUGU's Academic Schedules with Precision & AI

Ready to transform classroom coordination across Indian Higher Education.